

● MEDIUM

● PROBLEM-SOLVING AND DATA ANALYSIS

● ANSWER KEY

Problem-Solving and Data Analysis

05

10 answers · Rationale-rich · Teach from doc

Q1**D****D. 98**

Choice D is correct. It's given that at a movie theater, there are a total of 350 customers and that each customer is located in either theater A, theater B, or theater C. If the probability of selecting a customer in theater A is 0.48, then 0.48×350 , or 168, customers are located in theater A. If the probability of selecting a customer in theater B is 0.24, then 0.24×350 , or 84, customers are located in theater B. It follows that there are $168 + 84$, or 252, customers in theater A and theater B. Therefore, there are $350 - 252$, or 98, customers in theater C.

Choice A is incorrect. This is the percent, not the number, of the customers that are located in theater C.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the number of customers that are located in theater B, not theater C.

Q2**D**

D. $y = 9.4 - 0.9x$

Choice D is correct. The data points suggest that as the variable x increases, the variable y decreases, which implies that an appropriate linear model for the data has a negative slope. The data points also show that when x is close to 0, y is greater than 9. Therefore, the y -intercept of the graph of an appropriate linear model has a y -coordinate greater than 9. The graph of an equation of the form $y = a + bx$, where a and b are constants, has a y -intercept with a y -coordinate of a and has a slope of b . Of the given choices, only choice D represents a graph that has a negative slope, -0.9 , and a y -intercept with a y -coordinate greater than 9, 9.4 .

Choice A is incorrect. The graph of this equation has a positive slope, not a negative slope, and a y -intercept with a y -coordinate less than 1, not greater than 9.

Choice B is incorrect. The graph of this equation has a y -intercept with a y -coordinate less than 1, not greater than 9.

Choice C is incorrect. The graph of this equation has a positive slope, not a negative slope.

Q3**B**

B. Median

Choice B is correct. The median weight is found by ordering the horses' weights from least to greatest and then determining the middle value from this list of weights. Decreasing the value for the horse with the lowest weight doesn't affect the median since it's still the lowest value.

Choice A is incorrect. The mean is calculated by finding the sum of all the weights of the horses and then dividing by the number of horses. Decreasing one of the weights would decrease the sum and therefore decrease the mean. Choice C is incorrect. Range is the difference between the highest and lowest weights, so decreasing the lowest weight would increase the range.

Choice D is incorrect. Standard deviation is calculated based on the mean weight of the horses. Decreasing one of the weights decreases the mean and therefore would affect the standard deviation.

Q4**D**

D. 17,136.0

Choice D is correct. Since 1 minute = 60 seconds and 1 hour = 60 minutes, it follows that 1 hour = (60)(60), or 3,600 seconds. Using this conversion factor, the space station's average speed of 4.76 miles per second is equal to an average speed of

$$\frac{4.76 \text{ miles}}{\text{second}} \times \frac{3,600 \text{ seconds}}{\text{hour}} = \frac{17,136 \text{ miles}}{\text{hour}}, \text{ or } 17,136 \text{ miles per hour.}$$

Choice A is incorrect. This is the space station's average speed in miles per minute. Choice B is incorrect. This is double the space station's average speed in miles per minute, or the number of miles the space station travels on average in 2 minutes. Choice C is incorrect. This is triple the space station's average speed in miles per minute, or the number of miles the space station travels on average in 3 minutes.

Q5**A**

A. It is plausible that the proportion is between 0.45 and 0.53.

Choice A is correct. It's given that the estimate for the proportion of the population that has the characteristic is 0.49 with an associated margin of error of 0.04. Subtracting the margin of error from the estimate and adding the margin of error to the estimate gives an interval of plausible values for the true proportion of the population that has the characteristic. Therefore, it's plausible that the proportion of the population that has this characteristic is between 0.45 and 0.53.

Choice B is incorrect. A value less than 0.45 is outside the interval of plausible values for the proportion of the population that has the characteristic.

Choice C is incorrect. The value 0.49 is an estimate for the proportion based on this sample. However, since the margin of error for this estimate is known, the most appropriate conclusion is not that the proportion is exactly one value but instead lies in an interval of plausible values.

Choice D is incorrect. A value greater than 0.53 is outside the interval of plausible values for the proportion of the population that has the characteristic.

Q6**C**

C. 1,170,000

Choice C is correct. Of the 1,300,000 units sold during the first month, 15% were returned, so $(1,300,000)(0.15) = 195,000$ units were returned during the first month. If the units were sold and returned at the same rate for the next 5 months, then a total of $(195,000)(6) = 1,170,000$ smartphone units were returned during the 6-month period.

Choice A is incorrect. This is the number of units that were returned in 1 month. Choice B is incorrect. This is the number of units that were returned in 5 months. Choice D is incorrect. This is the number of units sold and not returned during the first 6 months.

Q7**B**B. $\frac{10}{34}$

Choice B is correct. The sample space is restricted to the presidents who were at least 50 years old when they first took office. Therefore, the sum of the values in the final four rows of the table, $13 + 11 + 7 + 3 = 34$, is the total number of presidents in the sample space. The number of presidents who were at least 60 years old is the sum of the values in the final two rows of the table: $7 + 3 = 10$. Thus, the fraction of the 34 presidents who were at least 50 years old when they first took office who were at least 60 years old is $\frac{10}{34}$.

Choice A is incorrect. This is the fraction of all presidents in the table who were at least 60 years old when they first took office. Choice C is incorrect and may result from treating the number of presidents who were between 50 and 59 years old when they first took office, instead of the number of presidents who were at least 50 years old, as the sample space. Choice D is incorrect and may result from a calculation error.

Q8

C

C. There is a constant c such that if $0 < x < c$, then $2^x < 2x + 2$, but if $x > c$, then $2^x > 2x + 2$.

Choice C is correct. At $x = 0$, the value of 2^x is less than the value of $2x + 2$: $2^0 < 2(0) + 2$, which is equivalent to $1 < 2$. As the value of x increases, the value of 2^x remains less than the value of $2x + 2$ until $x = 3$, which is when the two values are equal: $2^3 = 2(3) + 2$, which is equivalent to $8 = 8$. Then, for $x > 3$, the value of 2^x is greater than the value of $2x + 2$. So there is a constant, 3, such that when $0 < x < 3$, then $2^x < 2x + 2$, but when $x > 3$, then $2^x > 2x + 2$.

Choice A is incorrect because $2^x > 2x + 2$ when $x > 3$. Choice B is incorrect because $2^x < 2x + 2$ when $0 < x < 3$. Choice D is incorrect because $2^x < 2x + 2$ when $0 < x < 3$ and $2^x > 2x + 2$ when $x > 3$.

Q9**A**

- A. The standard deviation of number of units sold for company A is less than the standard deviation of number of units sold for company B.
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Choice A is correct. Standard deviation measures the spread of a given data set from its mean. In a data set with a smaller standard deviation, there are more values close to the mean. In a data set with a greater standard deviation, there are more values farther from the mean. The two histograms shown have the same scale on the horizontal axis. Therefore, their standard deviations can be compared by visually comparing the spreads of their histograms. The distribution summarized by each histogram is symmetric. Therefore, the mean of the data set for each histogram is a value in the middle bar of that histogram. The middle bar of each histogram has a value of at least 30 thousand units sold but less than 35 thousand units sold. Therefore, the mean of the data set for each histogram is at least 30 thousand and less than 35 thousand. The histogram for company A shows all the values in that data set are close to the mean. For company B, the histogram shows there are fewer values close to the mean and more values farther from the mean. Therefore, the standard deviation of number of units sold for company A is less than the standard deviation of number of units sold for company B.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q10**980**

The correct answer is 980. It's given that the ratio 140 to m is equivalent to the ratio 4 to 28. Therefore, the value of m can be found by solving the equation $\frac{140}{m} = \frac{4}{28}$. Multiplying each side of this equation by m yields $140 = \frac{4m}{28}$. Multiplying each side of this equation by 28 yields $3,920 = 4m$. Dividing each side of this equation by 4 yields $980 = m$. Therefore, the value of m is 980.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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